

Supply Chain Analysis – Python Workflow Summary

1. Data Loading & Environment Setup

- Imported Pandas, NumPy, Matplotlib and Seaborn for analysis and visualization.
- Applied visualization themes and loaded the supply chain dataset.

2. Exploratory Data Analysis (EDA)

- Checked dataset dimensions, duplicates, missing values and column structure.
- Reviewed categorical distributions and low-cardinality fields.

3. Data Cleaning

- Removed unnecessary and sensitive columns including customer details and image information.
- Prepared the dataset for operational analytics and modeling.

4. Feature Engineering

Created new metrics:

- Order Processing Time
- Delay Days
- Is_Delayed flag
- Month / Day / Hour variables
- Profitability categories (Profit / Loss / Break-even)

5. Business KPI Analysis

Measured delivery performance, late shipment percentage, profit impact and order-level metrics.

6. Profitability & Delay Analysis

- Profit distribution analysis
- Delay-day behavior study
- Profit vs delivery performance evaluation

7. Bottleneck Detection

Analyzed delay percentage across:

- Region
- Shipping Mode
- Department
- Customer Segment
- Payment Status

8. Root Cause Investigation

Performed focused analysis on high-delay regions and operational drivers affecting shipment performance.

9. Time-Series Pattern Analysis

Studied seasonal and temporal patterns using month, weekday and hourly trends.

10. Machine Learning Pipeline

- Frequency encoding
- Train-test split
- SMOTE balancing
- Random Forest classification
- Evaluation using Accuracy, Precision, Recall and F1-score

11. Business Outcomes

- Shipment risk prediction
- Delivery optimization opportunities
- Logistics bottleneck discovery
- Profit impact assessment

- Operational improvement recommendations